This section focuses on the theoretical foundations of the Finite Element Method. It is designed to help you develop a deeper understanding of the principles behind FEM and how they are applied in solving engineering problems.

This part of the coursework contributes **18% of the total module mark** (9% per question).

Instructions for Completing This Section

- You are required to **solve all given exercises by hand**. Typed or computer-generated solutions will **not be accepted**.
- After completing each exercise, **clearly photograph or scan your handwritten solutions**, making sure the question number and page order are visible.
- Include these photos in your **final report** in the same order as the questions appear.
- During the teaching sessions, various **solution techniques and example problems** will be introduced. You are encouraged to use these methods to guide your approach to solving the exercises in this section.

The questions for this section are provided in the following pages. Please read them carefully and ensure your solutions are well-organized and legible.

Question 1: (9% of the overall coursework mark)

The below figure shows a 2D truss structure consisting of **nine elements**. All elements have a Young's modulus of E=30 GPa and a cross-sectional area of A=20000 mm^2. Node 1 is supported by a **roller**, while node 6 is supported by a **pin**. Two vertical loads, P=(10+X) kN are applied at nodes 2 and 4, and a horizontal load W=(7.5+X) kN is applied at node 3. Here, **X is the last numerical digit of your P-number**.

Example: P----075 therefore X=5 and for instance the vertical loads are 15 (kN)
P----033x therefore X=3 and for instance the vertical loads are 13 (kN)

Task

You are required to **develop and solve a complete finite element model** of the structure and then critically analyse your results.

Your report must include the following sections:

Numerical Analysis (Required) (3 marks out of 9)

1. Develop a full FEM solution of the structure, including:
   - definition of the element type and degrees of freedom,
   - formulation and assembly of the global stiffness matrix,
   - application of boundary conditions and loads,
   - calculation of nodal displacements, reaction forces, element strains and stresses.

You may use analytical derivations, MATLAB, Python, or FEM software; however, **all modelling choices must be clearly stated and justified**.

Critical Analysis (Answer all parts below):

(a) Reconstruct the FEM model and clearly justify all modelling assumptions. **(1 mark out of 9)**

(b) *Without relying on a full numerical solution*, predict the **sign and relative magnitude** of nodal displacements. Your answer must explicitly refer to **at least one result from your own numerical solution** for validation. **(1 mark out of 9)**

(c) Identify the **most critical element in terms of stress** and explain why, using physical and structural reasoning rather than numerical ranking alone. **(1 mark out of 9)**

(d) Discuss how **uncertainties in boundary conditions** could affect the displacement and stress results. **(1 mark out of 9)**

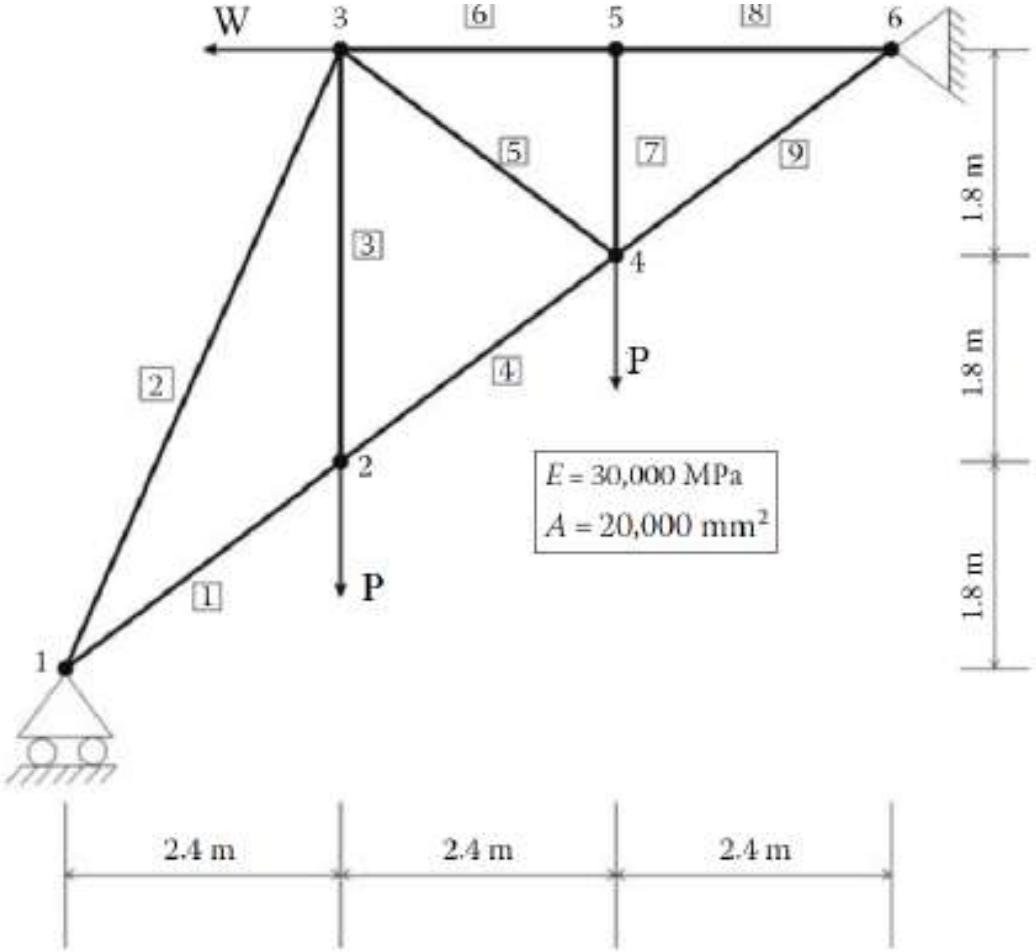
(e) Identify **one modelling assumption** whose violation would invalidate your FEM results, and justify your choice. **(0.75 mark out of 9)**

(f) Defend your selection of the most critical element under a **hypothetical 50% reduction of load P**, without re-solving the entire system numerically. **(0.75 mark out of 9)**

(g) State **which result you would trust the least** and explain why. **(0.5 mark out of 9)**

Important notes

- Numerical results alone will not receive full marks.
- Marks are awarded for **justification, reasoning, and engineering judgement**.
- Reports showing identical structure, wording, or reasoning patterns may be awarded different marks.



Question 2: (9% of the overall coursework mark)

The finite element model shown consists of four linear triangular elements. All modelling information is provided in the figure.

Important note: Not all modelling choices implied by the figure are physically or mathematically correct.

a) Boundary condition verification (2 marks)

Carefully examine the boundary conditions shown.

1. Identify **any incorrect or incomplete boundary condition**
2. Propose the correct condition
3. Explain the **consequence on stiffness, displacements, or reactions** if the incorrect condition is used

b) Stress state assessment (2 marks)

The problem may be modelled as either:

- Plane stress
- Plane strain

1. Select the correct assumption
2. Justify your choice **based on geometry and loading**
3. Quantify (qualitatively or approximately) how results would differ if the other assumption were used

(c) Load representation (2 marks)

The applied traction on the right boundary is uniform.

1. Derive the **equivalent nodal forces** for this boundary
2. Explain why a naive force lumping approach would be incorrect
3. State the impact on stress results if lumping is done improperly

(d) FEM solution and interpretation (3 marks)

Solve the problem using a **corrected and justified model**, and report:

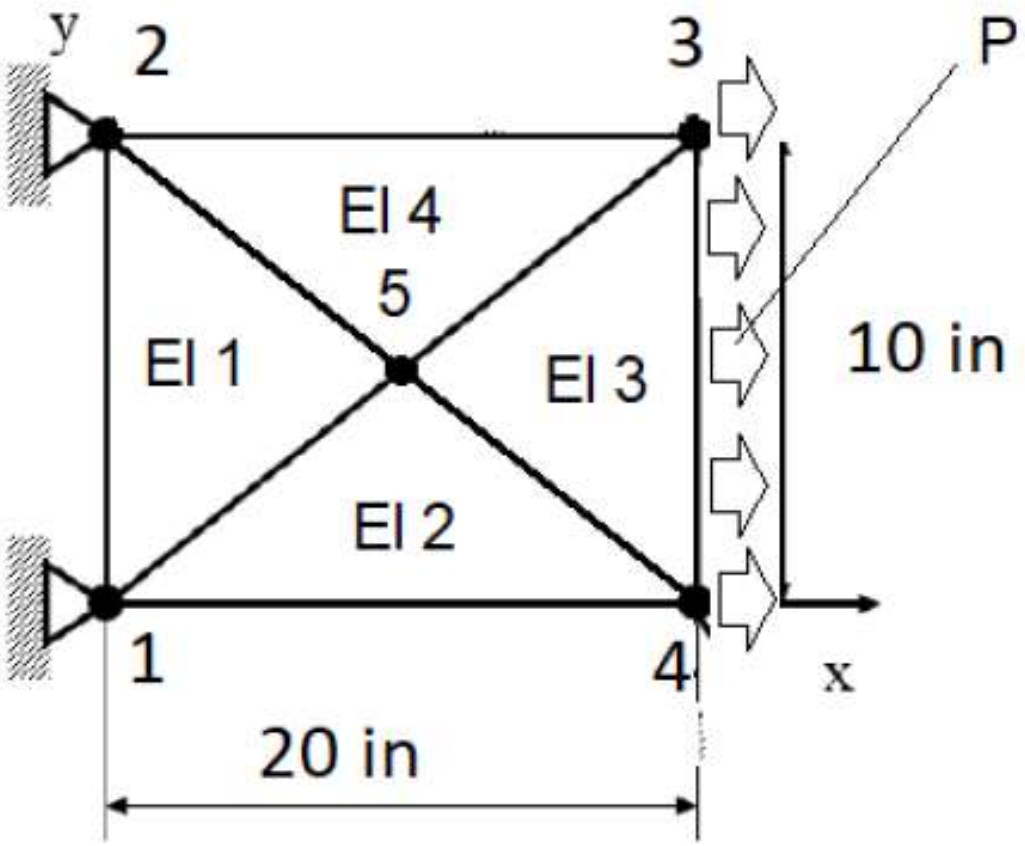
- Nodal displacements
- Reaction forces
- Element stresses

Finally, comment on:

- Whether the stress field is mesh-dependent
- Whether this mesh is adequate for stress prediction

Note: A load acting as P=1000+X psi where **X** represents the last 3 numerical digits of your P number.

For example:

- If your P number is P---175 therefore X=1175.
- If your P number is P---233x therefore X=1233.

The plate has dimensions of 20 inches long and 10 inches high. The thickness is 1 inch, the elastic modulus is 30×10^6 psi, and the poisons ratio is 0.3.